

Operating System as a Resource Manager

A computer has many resources (Hardware and Software), which may be required to complete a task. The commonly required resources are Input/Output devices, Memory file storage space, [CPU \(Central Processing Unit\)](#) time and so on.

The operating system acts as the manager of these resources and allocates them to specific programs and users as necessary for their tasks. Therefore, we can say an operating system is a **resource allocator**. This is the main features of an operating system.

When a number of computers connected through a network more than one computer trying for a computer print or a common resource, then the operating system follows same order and manage the resources in an efficient manner.

Generally, resources sharing in two ways "in time" and "in space". When a resource is a [time-sharing](#) resource, first one of the tasks get the resource for some time, then another and so on.

For example, a CPU in the time-sharing system. In the time-sharing system, OS fixes the time slot for the CPU. First one of the processes gets the CPU, when the time slot expires, the CPU switch to the next [process](#) in the ready queue. In this example, the CPU is a time resource. [CPU Scheduling in Operating System](#)

The other kind of sharing is the "**space sharing**". In this method, the users share the space of resource. For example, the main memory consisting of several processes at a time, so the process of sharing the resources.

The main difference between "in time" sharing resources and "in space" sharing resources is "in time" resources are not divided into units, whereas "in space" sharing resources are divided into units.

I hope you have cleared about the **Operating System** concepts.

CPU Scheduling Algorithms Problems With Solutions

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Process	Burst Time(ms)
P ₁	5
P ₂	24
P ₃	16
P ₄	10
P ₅	3

P ₁	P ₂	P ₃	P ₄	P ₅	
0	5	29	45	55	58

FCFS Scheduling

Today we will practice problems on different types of [CPU Scheduling Algorithms](#). We will see here that how CPU scheduler uses **scheduling algorithms** during execution of process. Let's see.

FCFS Example

Consider the above set of processes that arrive at time zero. The length of the CPU **burst time** given in millisecond. Now we calculate the average waiting time, average turnaround time and throughput.

Average Waiting Time and Turnaround Time

Average Waiting Time

First of all, we have to calculate the waiting time of each process.							
Waiting Time	=	Starting Time	-	Arrival Time			
Waiting time							
P ₁	=		=				0
P ₂	=	5	=	0	=	5	ms
P ₃	=	29	=	0	=	29	ms
P ₄	=	45	=	0	=	45	ms
P ₅	=	55	=	0	=	55	ms

Average Waiting Time = Waiting Time of all Processes / Total Number of Process
 Therefore, average waiting time = (0 + 5 + 29 + 45 + 55) / 5 = 25 ms

Average Turnaround Time

Turnaround Time = Waiting time in the ready queue + executing time + waiting time in waiting-queue for I/O

Turnaround time of

P1 = 0 + 5 + 0 = 5ms

P2 = 5 + 24 + 0 = 29ms

P3 = 29 + 16 + 0 = 45ms

P4 = 45 + 10 + 0 = 55ms

P5 = 55 + 3 + 0 = 58ms

Total Turnaround Time = (5 + 29 + 45 + 55 + 58) ms = 192ms

Average Turnaround Time = (Total Turnaround Time / Total Number of Process) = (192 / 5) ms = 38.4ms

Throughput

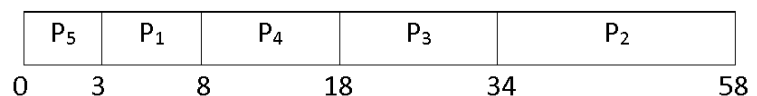
Here, we have a total of five processes. Process P1, P2, P3, P4, and P5 takes 5ms, 24ms, 16ms, 10ms, and 3ms to execute respectively.

Throughput = (5 + 24 + 16 + 10 + 3) / 5 = 11.6ms

It means one process executes in every 11.6 ms.

SJF (Shortest Job First) Scheduling

Process	Burst Time(ms)
P ₁	5
P ₂	24
P ₃	16
P ₄	10
P ₅	3



SJF Scheduling

Average Waiting Time and Turnaround Time

Average Waiting Time

We will apply the same formula to find average waiting time in this problem. Here arrival time is common to all processes(i.e., zero).

Waiting Time for

P1 = 3 - 0 = 3ms

P2 = 34 - 0 = 34ms

P3	=	18	-	0	=	18ms
P4	=	8	-	0	=	8ms
P5	=		=		=	0ms

Now, Average Waiting Time = $(3 + 34 + 18 + 8 + 0) / 5 = 12.6\text{ms}$

Average Turnaround Time

According to the SJF Gantt chart and the turnaround time formulae, Turnaround Time of

P1	=	3	+	5	=	8ms
P2	=	34	+	24	=	58ms
P3	=	18	+	16	=	34ms
P4	=	8	+	10	=	18ms
P5	=	0	+	3	=	3ms

Therefore, Average Turnaround Time = $(8 + 58 + 34 + 18 + 3) / 5 = 24.2\text{ms}$

Throughput

Here, we have a total of five processes. Process P1, P2, P3, P4, and P5 takes 5ms, 24ms, 16ms, 10ms, and 3ms to execute respectively. Therefore, Throughput will be same as above problem i.e., 11.6ms for each process.

Round Robin Scheduling Example

Here is the Round Robin scheduling example with gantt chart. Time Quantum is **5ms**.

Process	CPU Burst Time
P ₁	30
P ₂	6
P ₃	8

Gantt Chart

P ₁	P ₂	P ₃	P ₁	P ₂	P ₃	P ₁	P ₁	P ₁	P ₁	
0	5	10	15	20	21	24	29	34	39	44

Round Robin Scheduling

Average Waiting Time and Turnaround Time

Average Waiting Time

For finding Average Waiting Time, we have to find out the waiting time of each process.

Waiting Time of

P1 = 0 + (15 - 5) + (24 - 20) = 14ms

P2 = 5 + (20 - 10) = 15ms

P3 = 10 + (21 - 15) = 16ms

Therefore, Average Waiting Time = $(14 + 15 + 16) / 3 = 15\text{ms}$

Average Turnaround Time

Same concept for finding the Turnaround Time.

Turnaround Time of

P1 = 14 + 30 = 44ms

P2 = 15 + 6 = 21ms

P3 = 16 + 8 = 24ms

Therefore, Average Turnaround Time = $(44 + 21 + 24) / 3 = 29.66\text{ms}$

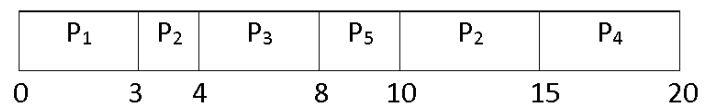
Throughput

Throughput = $(30 + 6 + 8) / 3 = 14.66\text{ms}$

In 14.66ms, one process executes.

Shortest Remaining Time First(SRTF)

Process	Burst Time(CPU)	Arrival Time(ms)
P ₁	3	0
P ₂	6	2
P ₃	4	4
P ₄	5	6
P ₅	2	8



SRTF Scheduling

Average Waiting Time and Turnaround Time

Average Waiting Time

First of all, we have to find the waiting time for each process.

Waiting Time of

P1 = 0ms

$$\begin{aligned}
 P2 &= (3 - 2) + (10 - 4) = 7\text{ms} \\
 P3 &= (4 - 4) = 0\text{ms} \\
 P4 &= (15 - 6) = 9\text{ms} \\
 P5 &= (8 - 8) = 0\text{ms}
 \end{aligned}$$

Therefore, Average Waiting Time = $(0 + 7 + 0 + 9 + 0) / 5 = 3.2\text{ms}$

Average Turnaround Time

First of all, we have to find the turnaround time of each process.

$$\begin{aligned}
 \text{Turnaround Time} & \\
 P1 &= (0 + 3) = 3\text{ms} \\
 P2 &= (7 + 6) = 13\text{ms} \\
 P3 &= (0 + 4) = 4\text{ms} \\
 P4 &= (9 + 5) = 14\text{ms} \\
 P5 &= (0 + 2) = 2\text{ms}
 \end{aligned}$$

Therefore, Average Turnaround Time = $(3 + 13 + 4 + 14 + 2) / 5 = 7.2\text{ms}$

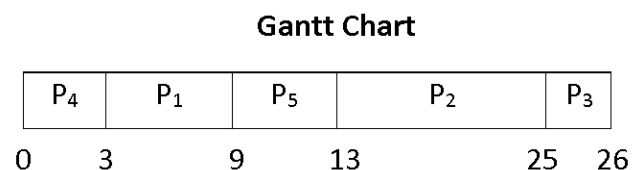
Throughput

$$\text{Throughput} = (3 + 6 + 4 + 5 + 2) / 5 = 4\text{ms}$$

Therefore, each process takes 4ms to execute.

Priority Scheduling Example

Process	CPU Burst Time	Priority
P ₁	6	2
P ₂	12	4
P ₃	1	5
P ₄	3	1
P ₅	4	3



Priority Scheduling Example with Gantt Chart

Average Waiting Time and Turnaround Time

Average Waiting Time

First of all, we have to find out the waiting time of each process.

Waiting Time of process

P1	=	3ms
P2	=	13ms
P3	=	25ms
P4	=	0ms
P5	=	9ms

Therefore, Average Waiting Time = $(3 + 13 + 25 + 0 + 9) / 5 = 10\text{ms}$

Average Turnaround Time

First finding Turnaround Time	Time of each process.
P1 = (3 + 6) = 9ms	
P2 = (13 + 12) = 25ms	
P3 = (25 + 1) = 26ms	
P4 = (0 + 3) = 3ms	
P5 = (9 + 4) = 13ms	

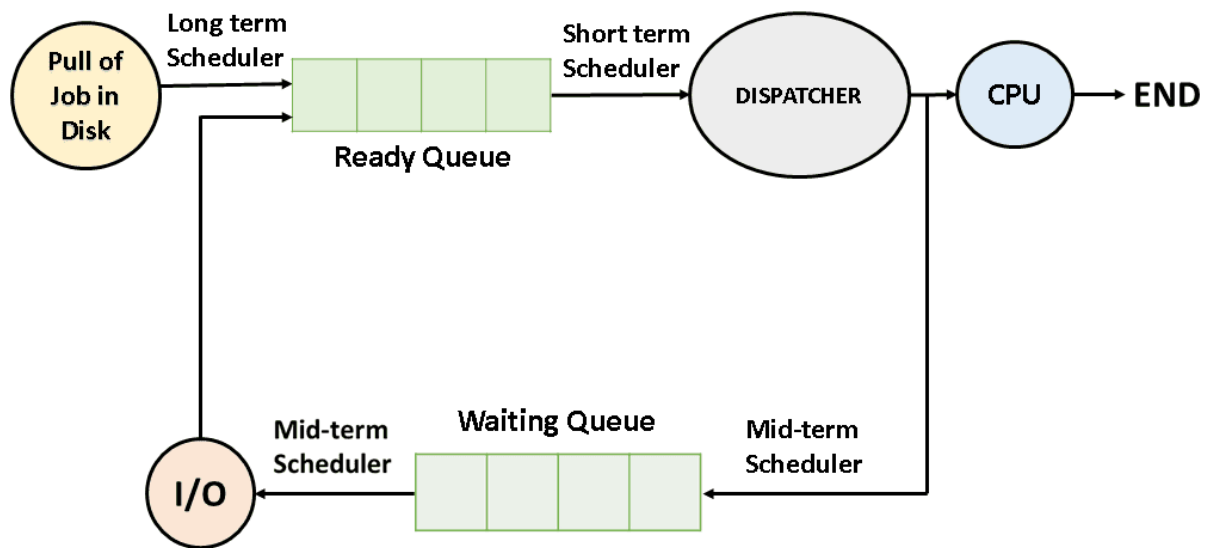
Therefore, Average Turnaround Time = $(9 + 25 + 26 + 3 + 13) / 5 = 15.2\text{ms}$

Throughput

Throughput = $(6 + 12 + 1 + 3 + 4) / 5 = 5.2\text{ms}$
 Therefore, each process takes 5.2ms to execute.

CPU Scheduling

CPU Scheduling in operating system is a fundamental function, almost all computer resources scheduled before use. The CPU is also one of the primary resources. So, CPU scheduling before use. The Scheduling Algorithms determine how the CPU will be allocated to the process.



Created by NotesJam

CPU Scheduling

CPU Scheduling in Operating System

CPU scheduling in the operating system is the basis of the multi-programmed operating system. The idea of multiprogramming is relatively simple if a process(job) is waiting for an I/O request, then the **CPU** switches from the process to another through a **dispatcher**, so the CPU is always busy in multiprogramming.

But in a simple computer system, the CPU sits idle until the **I/O request** granted. By switching the CPU among process, the OS can make the computer more productive.

Important

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Types of CPU Scheduling

CPU scheduling has two types-Non-Preemptive and Preemptive

Non-Preemptive Scheduling

In the non-preemptive scheduling, once the CPU assigned to a process, the processor does not release until the completion of that process. The CPU will be assigned to another job only after the previous has finished.

Preemptive Scheduling

In the preemptive scheduling algorithm, the CPU can release the process even in the middle of the execution. For example, the CPU executing the process P1, in the middle of the execution the CPU receives a request signal from process P2. Then the OS compares the priorities of P1 and P2.

CPU Scheduler in OS

CPU schedulers in operating system are:

Long-term Scheduler

The function of a long-term scheduler is to select the jobs from the pool of jobs and load these jobs into main memory(Ready Queue) of that computer, so the long-term scheduler is also called **job-scheduler**.

Short-term Scheduler

The function of a short-term scheduler is to select a job from the ready queue and gives the control of the CPU to that process with the help of "Dispatcher". That's why the short-term scheduler is also said to be the **CPU scheduler**.

The method of selecting a process from the ready queue by the short-term scheduler is depending on the [Scheduling algorithms](#).

Medium-term Scheduler

If a process requests an I/O in the middle of execution, then the process removed from the main memory is loaded into the waiting queue. When the I/O operation is completed, then the job moved from waiting queue to ready queue. These two operations are performed by the medium-term scheduler.

Dispatcher in OS

A dispatcher is a module, it connects the CPU to the process selected by the short-term scheduler. The main function of the dispatcher is **switching**, it means switching the CPU from one process to another process.

Another function of the dispatcher is jumping to the proper location in the user program and ready to start execution. The time taken by the dispatcher to stop one process and start another running is called the **Dispatch Latency** in OS.

The degree of multiprogramming is dependent on the dispatch latency. If the dispatch latency is increasing, then the degree of multiprogramming decreases.

CPU Scheduling Criteria in OS

Throughput in OS

It means how many jobs are completed by the CPU.

Turnaround Time in OS

The time interval between the submission of the process and the time of the completion is the turnaround time.

Important

Turnaround time = Waiting time in the ready queue + executing time + waiting time in waiting queue for I/O.

Waiting Time in OS

Waiting time is the sum of the periods spent waiting by a process in the ready queue.

Response Time in OS

Response time is the time duration between the submission and the first response.

CPU Utilization in OS

This is the percentage of time that the processor is busy, CPU utilization may range from **0-100%**.

I hope you have cleared about **CPU Scheduling in OS**, you also learned types of scheduling, Scheduling criteria, dispatcher and three kinds of schedulers i.e., long term, short term, and medium-term scheduler.